GROSS ANATOMY · TBL GROUP 17 · CASE 4 (GROUP 4)

The Rugby-Tackle Shoulder: Anterior Dislocation and a Stretched Lateral Cord

A beginner's walkthrough to prepare for the case discussion

How to use this handout

- Read Sections 1–4 to understand the story of this patient.
- Use Section 5 to answer each objective aloud.
- Use Section 6 to practise the follow-up questions.
- Do the self-check in Section 7 before you look at the answers.

WHAT I WORKED FROM

Attached: the case scenario, the five learning objectives, and your group's task breakdown.

Not attached: a separate guide-question list and lecture materials. So Section 5 uses each objective, with your group's assigned focus points as its sub-parts. No lecture mnemonics appear. The three memory aids in Section 7 are common study aids, not from your lecture.

1. The case in plain language
2. What I need to know first
3. What changes in this patient — and why
4. Understanding the important findings
5. Every objective answered
6. Likely professor follow-up questions
7. Quick recap and self-check
8. Appendix: source check and references

1. The case in plain language

A **22-year-old rugby player** is tackled hard from the side. At that moment his arm is stretched out and turned outward. He lands on the ground with **severe pain in his shoulder**.

In the emergency room, the doctors find that the ball at the top of his upper arm bone has slipped out of the shoulder socket and moved forward. This is an **anterior dislocation of the glenohumeral joint**. *Anterior* means "toward the front". The *glenohumeral joint* is the shoulder joint, the joint between the arm bone and the shoulder blade.

The ball did not only leave its socket. It moved into the space where a large bundle of arm nerves runs. This bundle is the **brachial plexus**. The displaced ball pushed on one part of the bundle, the **lateral cord**, and stretched it hard. The stretch damaged the nerve that the lateral cord turns into as it continues down the arm (its *terminal branch*).

THE CENTRAL PROBLEM

How can an injury to a joint damage a nerve — and what will this patient be unable to do because of it?

Item	What we know
STATED Diagnosis	Anterior dislocation of the glenohumeral (shoulder) joint.
STATED Nerve injury	Extreme stretch of the lateral cord of the brachial plexus, which damaged "its terminal branch".
MOST LIKELY Which branch	The **musculocutaneous nerve**. The case does not name it. Section 3 (Step 4) explains the reasoning.
ALSO POSSIBLE	The **lateral root of the median nerve**, the lateral cord's other terminal branch.

Item	What we know
NOT MENTIONED	The **axillary nerve**. In real patients it is the nerve most often hurt in an anterior dislocation, but it comes from a different cord. A careful student says it should be checked.

What the objectives ask me to understand about this patient

Learning objective	The question it asks about this patient
1. Terminal branches of the brachial plexus	What are the five end-nerves, and which of them come from the lateral cord?
2. Parts of the humerus in plexus injuries	Which part of the bone moved (the head)? Which nearby part carries a nerve (the surgical neck)?
3. Muscles supplied by the terminal branches	Which muscles lose their nerve supply in this patient?
4. Actions affected	Which movements become weak, and which stay normal?
5. Deficits of each nerve injury	What would the examination show for this nerve, and how does it differ from the other four?

2. What I need to know first

This section teaches only the normal anatomy that the case needs. Each idea builds on the one before it.

2.1 The shoulder joint — a large ball on a shallow socket

What it is. The **glenohumeral joint** is the joint between the **head of the humerus** (the ball at the top of the upper arm bone) and the **glenoid cavity** (the shallow socket on the side of the **scapula**, or shoulder blade).

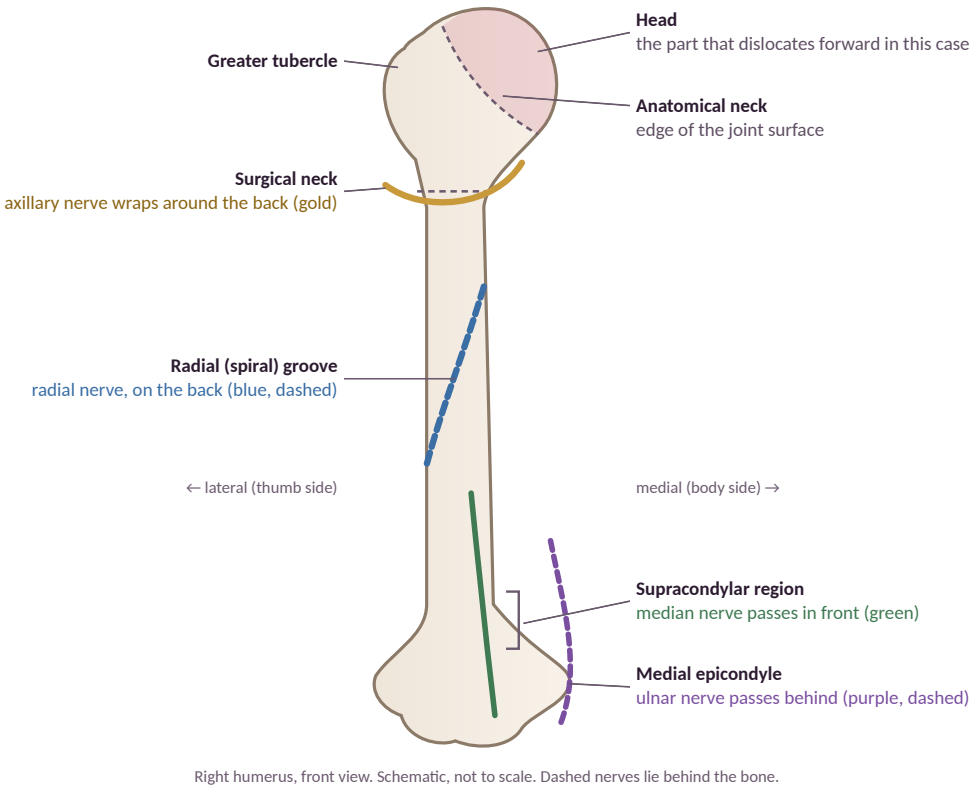
Normal job. It lets the arm move in almost every direction. It is the most mobile joint in the body.

Why it matters here. The ball is much larger than the socket. That gives mobility, but it costs stability. Soft tissues hold the ball in place: the **capsule** (a tough sleeve around the joint), the **labrum** (a rim of cartilage that deepens the socket), and the surrounding muscles. When the arm is lifted out to the side and turned outward, the front-lower part of the capsule is the weakest support. That is the exact position in this case. A **dislocation** means that the two joint surfaces lose contact completely.

2.2 The humerus — the parts that matter for nerves

Part	Where it is	Why it matters in this case
Head	The smooth, rounded ball at the top. It faces inward toward the socket.	This is the part that dislocates and pushes on the nerves.
Anatomical neck	The narrow groove right at the edge of the head.	Marks the edge of the joint surface. Do not confuse it with the surgical neck.
Greater and lesser tubercles	Two bumps beside the head. Rotator cuff muscles attach to them.	Landmarks. The surgical neck lies just below them.
Surgical neck	The narrow part just below the tubercles, where the top of the bone meets the shaft.	A common fracture site. The **axillary nerve** wraps around the back of it.
Radial (spiral) groove	A shallow groove that spirals across the back of the shaft.	The **radial nerve** lies in it.

Part	Where it is	Why it matters in this case
Supracondylar region	The lower shaft, just above the elbow end of the bone.	The **median nerve** and the brachial artery pass in front of it.
Medial epicondyle	The bump on the inner side of the elbow end.	The **ulnar nerve** passes behind it (the "funny bone").



Right humerus, front view. Schematic, not to scale. Dashed nerves lie behind the bone.

Figure 1. The humerus and its four nerve "contact points". The surgical neck carries the axillary nerve, the radial groove carries the radial nerve, the median nerve passes in front of the supracondylar region, and the ulnar nerve passes behind the medial epicondyle. The musculocutaneous nerve does not touch the bone. It runs inside the muscles in front of it.

2.3 Nerves — cables that carry signals both ways

A **nerve** is a cable made of many **nerve fibres** bundled together. **Motor fibres** carry commands from the spinal cord to muscles. **Sensory fibres** carry touch, pain, and position signals from the skin back to the spinal cord. When a nerve **innervates** a muscle or a patch of skin, it supplies it with these fibres.

Why it matters here. A damaged nerve causes two kinds of loss: **weak muscles** (motor loss) and **numb skin** (sensory loss). Each nerve has its own set of muscles and its own skin patch. So the pattern of loss tells you which nerve is damaged.

2.4 The brachial plexus — the wiring box of the upper limb

What it is. The **brachial plexus** is a network of nerves formed by the front branches (*anterior rami*) of five spinal nerves: **C5, C6, C7, C8, and T1**. It supplies the skin and muscles of the whole upper limb. A *plexus* is a network where nerve fibres mix and regroup.

Its five parts, in order from the neck to the arm:

1. **Roots** — the five spinal nerves C5 to T1.

2. **Trunks** — three. Upper (C5 + C6), middle (C7), and lower (C8 + T1).
3. **Divisions** — each trunk splits into an **anterior** and a **posterior** division. Anterior divisions mainly feed the front (flexor) muscles. Posterior divisions feed the back (extensor) muscles.
4. **Cords** — three. **Lateral, posterior, and medial**. They are named by where they sit around the **axillary artery** (the main artery of the armpit), behind the **pectoralis minor** muscle.
5. **Terminal branches** — the five nerves that the cords end as.

Why it matters here. The cords run just in front of and below the shoulder joint, close to the **coracoid process**. The coracoid is a hook of bone that sticks forward from the scapula. A humeral head that moves forward and down lands right against the cords.

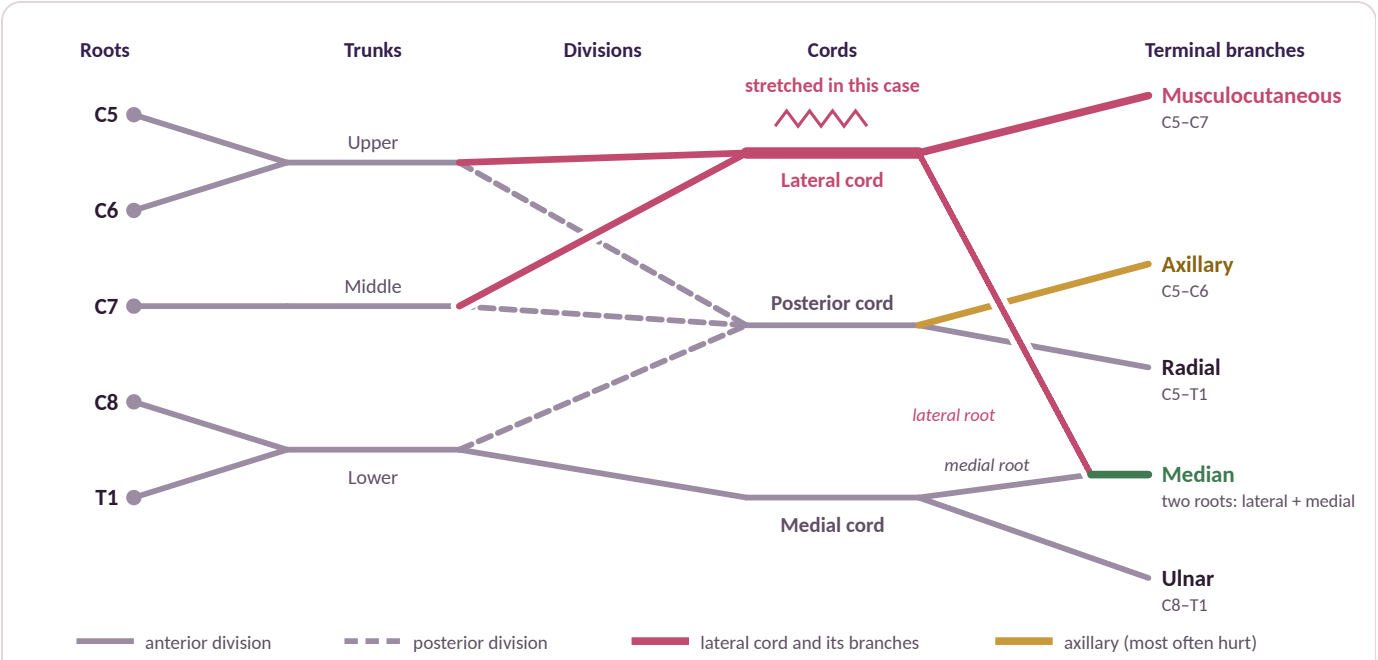


Figure 2. The brachial plexus, simplified. The lateral cord (pink) forms from the anterior divisions of the upper and middle trunks. It ends as two terminal branches: the musculocutaneous nerve and the lateral root of the median nerve. The median nerve is the only terminal branch with two roots. Where lines cross, the white gap shows one passing over the other. Schematic only.

2.5 The lateral cord and its branches

- **How it forms:** from the anterior divisions of the upper and middle trunks. So it carries fibres from **C5, C6, and C7**.
- **Terminal branch 1 — the musculocutaneous nerve.** A complete, named nerve of its own.
- **Terminal branch 2 — the lateral root of the median nerve.** It is only half of a nerve. It joins the medial root (from the medial cord) to make the median nerve.
- **A side branch:** the lateral pectoral nerve (to pectoralis major). It leaves the cord along the way, so it is a *collateral* branch, not a terminal one.
- **The pattern to remember:** the lateral cord is built from *anterior* divisions. So its nerves supply *front* muscles — the elbow flexors and some forearm flexors.

2.6 Movement words you will need (Objective 4)

Term	Meaning	Everyday example
Flexion (elbow)	Bending: the forearm moves toward the arm.	Lifting a cup to your mouth
Extension (elbow)	Straightening.	Pushing a door open
Abduction (shoulder)	Lifting the arm out to the side, away from the body.	The "up" part of a jumping jack
Adduction	Bringing the arm back toward the body.	Holding a book under your arm
Lateral (external) rotation	Turning the arm outward.	The position in this case
Medial (internal) rotation	Turning the arm inward.	Reaching behind your back
Supination	Turning the forearm so the palm faces up.	Holding a bowl of soup; turning a screwdriver to tighten (right hand)
Pronation	Turning the forearm so the palm faces down.	Typing on a keyboard
Grip	Closing the fingers around an object. It needs finger flexors *and* a wrist held slightly extended.	Holding a bag handle

3. What changes in this patient — and why

Here is the full chain: **tackle → head leaves the socket → head presses on the cords → stretch damages nerve fibres → muscles and skin lose their nerve → the patient's deficits.** Each step is explained below.

1

The tackle levers the ball out of the socket

His arm was out to the side, turned outward, and probably pulled back. In this position the front-lower part of the capsule is tight. The head of the humerus already presses against the front of the socket. The hard hit from the side adds force to this lever. The capsule and labrum tear at the front, and the head slips over the front-lower rim of the glenoid. It usually comes to rest in front of and below the socket, under the coracoid process. This is called the **subcoracoid** position. Because this position is so common in sport, anterior dislocation is by far the most common type of shoulder dislocation.

Links to the case: severe shoulder pain; the ER diagnosis of anterior dislocation.

Side effect on the bone: while the head sits out of place, its soft back-outer surface can be pressed against the hard rim of the socket. This leaves a dent called a **Hill–Sachs lesion**. The case does not say whether this happened.

2

The displaced head pushes into the nerve bundle

The cords lie just in front of the shoulder joint, behind pectoralis minor, near the coracoid. The head now sits in that same space. It pushes the cords forward. A cord is anchored above (in the neck) and below (where its branches enter muscles). When something pushes on the middle of an anchored cord, the cord stretches. Think of a tight rope that you press in the middle — the rope gets longer and tighter between its two fixed ends.

Links to the case: "The forward displacement of the humerus head puts an extreme stretch on the lateral cord."

3 The stretch damages the nerve fibres

Nerve fibres tolerate very little stretch. A strong stretch can stop the fibres from carrying signals for a while, even though the nerve looks intact. A stronger stretch can tear fibres inside the nerve. Either way, the signal stops. Commands no longer reach the muscles, and touch signals from the skin no longer reach the spinal cord. The case does not tell us how severe the damage is.

4 Why the musculocutaneous nerve is the most likely "terminal branch"

- The lateral cord ends as two branches. Only one of them is a complete, named nerve: the **musculocutaneous nerve**. The other is one root of the median nerve. The case uses the singular "its terminal branch".
- The musculocutaneous nerve leaves the lateral cord near the coracoid. It then passes *through* the coracobrachialis muscle, where it is held fairly fixed. A head pushing forward stretches the nerve against this fixed point. PROPOSED MECHANISM — case reports suggest it; it is not proven.
- **Conclusion:** the musculocutaneous nerve is the best answer. The lateral root of the median nerve is also at risk, and both can be hurt together.

5 From a damaged nerve to this patient's deficits

Motor. The musculocutaneous nerve is damaged, so signals no longer reach its three muscles: **coracobrachialis, biceps brachii, and brachialis**. These muscles cannot contract with normal force. As a result, **elbow flexion becomes weak**, **supination becomes weak** (the biceps is the strongest supinator), and shoulder flexion becomes slightly weak. Over the next weeks, the front of the arm thins out (**atrophy**), because a muscle without its nerve supply shrinks.

Sensory. The nerve continues past the elbow as the **lateral cutaneous nerve of the forearm**. With that nerve silent, a strip of skin on the **thumb side of the forearm** becomes numb or tingly.

Reflex. A **reflex** is an automatic muscle response to a tap. The **biceps reflex** (tap the biceps tendon, the biceps twitches) travels through this nerve and spinal levels C5–C6. With the nerve damaged, the reflex becomes weak or absent.

WHY WEAK — AND NOT PARALYSED?

Other muscles can still bend the elbow. The **brachioradialis** is supplied by the **radial nerve**. It flexes the elbow best with the thumb pointing up. In many people, a small radial nerve branch also reaches the outer part of the brachialis. So the patient can still bend the elbow, but weakly. In the same way, supination remains possible through the **supinator** muscle (radial nerve), but it is weak — most of all when the elbow is bent, because that is when the biceps normally does most of the work.

6 If the lateral root of the median nerve is also damaged POSSIBLE, NOT STATED

The lateral root carries mainly C6–C7 fibres. Most of its motor fibres go to forearm muscles such as **pronator teres** (pronation) and **flexor carpi radialis** (wrist flexion toward the thumb side). It probably also carries sensory fibres for the thumb side of the hand. So you might see **weaker pronation**, **weaker wrist flexion**, and possibly numbness on the thumb side of the palm and fingers.

You would **not** expect wasting of the thumb muscles (the **thenar** muscles). Their fibres (C8–T1) come through the medial root, which is intact.

7 A nerve the case does not mention: the axillary nerve

The **axillary nerve** comes from the posterior cord. It wraps around the surgical neck of the humerus. A head that drops forward and down drags on it. In real patients it is the nerve most often injured in anterior dislocation. Doctors therefore test deltoid strength and the skin over the lower deltoid before and after they put the joint back in place (**reduction**). If it were also injured, the patient would have weak abduction, weak lateral rotation, and a numb patch on the outer shoulder.

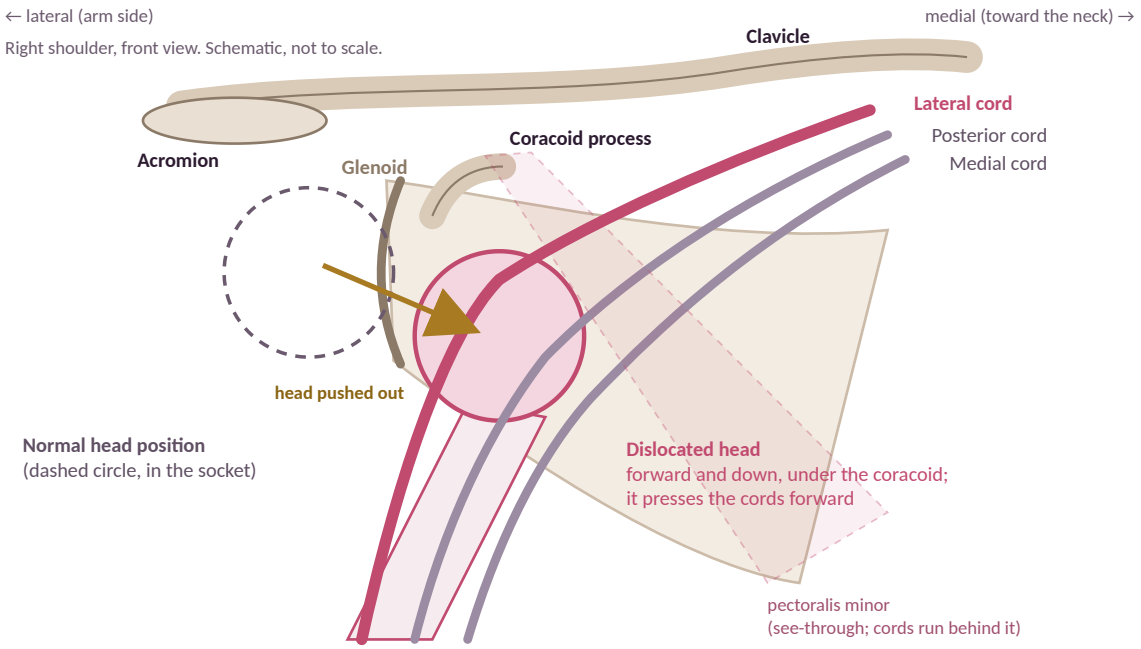


Figure 3. Where the dislocated head ends up. In a front view, "forward" points toward you, so the displaced head appears below and inside the socket, under the coracoid. That is the same space the cords pass through on their way into the arm. The cord closest to the head and the coracoid — the lateral cord — takes the stretch in this case.

Stated, supported, and uncertain

Category	Details
STATED IN THE CASE	22-year-old rugby player; tackled from the side; arm fully extended and rotated outward; severe shoulder pain; anterior dislocation of the glenohumeral joint; extreme stretch of the lateral cord; damage to its terminal branch.
SUPPORTED BY REASONING	The damaged branch is most likely the musculocutaneous nerve. Expected deficits: weak elbow flexion and supination, a weak biceps reflex, and numbness on the thumb side of the forearm.
UNCERTAIN	Whether the lateral root of the median nerve is also damaged. How severe the nerve injury is. Whether the axillary nerve is also hurt. Whether there is a fracture (no imaging is reported). The axillary artery runs with the cords; the case does not report on it, so a pulse check would be routine.

4. Understanding the important findings

The case gives no laboratory values and no imaging numbers. So no reference ranges apply. Rows marked EXPECTED are findings you would look for on examination. The case does **not** report them.

Case finding	What it means	Why it matters in this case
STATED IN THE CASE		
22-year-old male rugby player	A young, active person in a contact sport.	Anterior dislocation is most common in young people (about 15–30 years) and more common in males. He fits the classic picture.
Tackled hard from the side	A large outside force on the arm.	The force drives the lever that pushes the head out of the socket.
Arm "fully extended and rotated outward"	In anatomy terms: abducted (out to the side), laterally rotated, and probably extended.	This is the classic position for anterior dislocation. The front-lower capsule is under the most strain.
Hit the ground; severe shoulder pain	A joint injury with torn capsule and stretched soft tissue.	Pain fits a dislocation, but it does not tell you which nerve is hurt. Pain also limits effort, so strength testing is repeated after reduction.
ER diagnosis: anterior dislocation	The head moved forward (and usually down, under the coracoid).	This places the head right next to the cords.
Forward displacement stretched the lateral cord	A **traction** (stretch) injury of the nerve.	Explains how a joint injury becomes a nerve injury.
Damage to "its terminal branch"	Most likely the musculocutaneous nerve.	Predicts every deficit listed below.
EXPECTED FINDINGS THAT WOULD SUPPORT A MUSCULOCUTANEOUS INJURY		
EXPECTED Weak elbow flexion, worst with the palm up	Biceps and brachialis are weak.	Supports the diagnosis. It is weakness, not total loss, because brachioradialis (radial nerve) still works.
EXPECTED Weak supination with the elbow bent	The biceps is weak.	Supports the diagnosis. The supinator muscle keeps some supination.

Case finding	What it means	Why it matters in this case
EXPECTED Weak or absent biceps reflex	The reflex loop through the nerve is broken.	Points to a nerve problem rather than a pure muscle problem.
EXPECTED Numb strip on the thumb side of the forearm	The lateral cutaneous nerve of the forearm is silent.	The strongest single clue that a *nerve* is damaged. A torn biceps tendon makes the elbow weak but leaves the skin normal.
EXPECTED LATER Thin front of the arm	Atrophy of the denervated muscles.	Takes weeks to appear. It would not be seen in the ER.
EXPECTED NORMAL FINDINGS THAT HELP		
EXPECTED Normal deltoid strength and normal skin over the lower deltoid	The axillary nerve works.	Rules out the most common nerve injury of dislocation.
EXPECTED Normal wrist and finger extension	The radial nerve works.	Shows the posterior cord is spared.
EXPECTED Normal finger spreading and thumb opposition	The ulnar nerve and the medial-root fibres of the median nerve work.	Shows the damage stays in the lateral cord territory.

Imaging: an X-ray would normally confirm the direction of the dislocation and look for a fracture. The case does not report one, so we cannot say whether a Hill–Sachs lesion or another fracture is present.

5. Every objective answered

No separate guide-question list was attached. Each learning objective below is treated as a guide question, keeping its original number. Your group's assigned focus points are the sub-parts. Each question is reproduced before its answer.

OBJECTIVE 1 · MEMBER 1

"To identify the terminal branches of the brachial plexus."

Focus: "List and briefly trace the five major terminal branches (Musculocutaneous, Axillary, Radial, Median, Ulnar), noting their cord origins, with a special spotlight on the lateral cord components highlighted in the scenario."

ANSWER I CAN SAY ALOUD

The brachial plexus ends as five terminal branches: the musculocutaneous, axillary, radial, median, and ulnar nerves. The lateral cord gives the musculocutaneous nerve and the lateral root of the median nerve. The posterior cord gives the axillary and radial nerves, and the medial cord gives the ulnar nerve and the medial root of the median nerve. In our patient, the stretched lateral cord puts the musculocutaneous nerve — and possibly the lateral root of the median — at risk.

Why this is the answer

The cords are the last stage of the plexus. Each cord splits into its end-nerves. The median nerve is special: it gets one root from the lateral cord and one from the medial cord. The two roots join in front of the axillary artery.

Nerve	Cord	Spinal levels	Short course	Main job
Musculocutaneous	Lateral	C5–C7	Pierces coracobrachialis; runs between biceps and brachialis; ends as the lateral cutaneous nerve of the forearm.	Front-of-arm muscles; skin of the thumb side of the forearm.
Median	Lateral + medial	C6–T1 (often C5)	Two roots join in the axilla; runs down the arm with no branches; crosses the front of the elbow; enters the hand through the carpal tunnel.	Most forearm flexors and both pronators; thumb (thenar) muscles; skin of the thumb-side palm and 3½ fingers.
Ulnar	Medial	C8–T1 (often C7)	Down the inner arm; behind the medial epicondyle; into the hand.	1½ forearm flexors; most small hand muscles; skin of the little finger and half the ring finger.
Axillary	Posterior	C5–C6	Through the quadrangular space; wraps around the surgical neck.	Deltoid and teres minor; skin over the lower deltoid.
Radial	Posterior	C5–T1	Runs in the radial groove on the back of the humerus; splits near the elbow into a skin branch and a muscle branch.	All back-of-arm and back-of-forearm (extensor) muscles; skin of the back of the arm, forearm, and thumb-side hand.

NOTE ON YOUR GROUP PLAN

The Objective 5 breakdown labels Member 7's part "Anterior & Lateral Cord Deficits". There is no anterior cord. The plexus has anterior *divisions* and three cords (lateral, medial, posterior). Also, the axillary nerve on Member 7's list comes from the *posterior* cord. A safer label: Member 7 — "Musculocutaneous, median, and axillary nerves"; Member 8 — "Radial and ulnar nerves, plus the sensory map".

OBJECTIVE 2 · MEMBER 2

"To identify the parts of the humerus involved in brachial plexus injuries."

Focus: "Discuss the head of the humerus and the surgical neck, explaining how an anterior dislocation forces the humeral head forward against the surrounding neurovascular structures, putting tension on the brachial plexus cords."

ANSWER I CAN SAY ALOUD

In this case the key part is the head of the humerus. It slipped out of the socket and moved forward and down to sit under the coracoid, which is exactly where the cords run, so it pushed them forward and stretched the lateral cord. The second key part is the surgical neck, the narrow area below the head where the axillary nerve wraps around the bone; that nerve is the one most often hurt in dislocations and in fractures there. Lower on the bone, the radial groove, the supracondylar region, and the medial epicondyle are where the radial, median, and ulnar nerves are at risk.

Why this is the answer

(a) The head and the cords. *Neurovascular structures* means nerves and blood vessels that travel together — here, the three cords and the axillary artery. They lie in front of the joint, with the subscapularis muscle between

them and the joint capsule. When the head moves forward, it pushes the subscapularis and the cords forward. The cords are anchored above and below, so the push turns into tension (stretch). The tension is greatest where a nerve is held fixed, such as where the musculocutaneous nerve enters coracobrachialis.

(b) Surgical neck versus anatomical neck. The **anatomical neck** is the groove at the edge of the head. The **surgical neck** is lower, below the tubercles, where the bone narrows into the shaft. The axillary nerve and the posterior circumflex humeral artery pass through the **quadrangular space** (a gap between muscles behind the shoulder) and wrap around the surgical neck. When the head drops forward and down, the axillary nerve is pulled tight around this point.

(c) Damage to the head itself. A **Hill–Sachs lesion** is a dent on the back-outer part of the head. It forms when the head is pressed against the rim of the socket during an anterior dislocation. Do not mix it up with a **Bankart lesion**: that is a tear of the front-lower labrum on the *scapula*, not on the humerus.

(d) The full humerus–nerve map (the wider objective):

Part of humerus	Nerve at risk	Typical injury	Main sign
Head	Cords of the plexus	Anterior dislocation (this case)	Depends on the cord stretched
Surgical neck	Axillary	Fracture; anterior dislocation	Weak abduction; numb skin over the lower deltoid
Radial groove (mid-shaft)	Radial	Shaft fracture	Wrist drop
Supracondylar region	Median	Supracondylar fracture (common in children)	Weak thumb and index flexion; numb thumb-side palm
Medial epicondyle	Ulnar	Fracture or pressure at the elbow	Clawing of the ring and little fingers; numb little finger

The musculocutaneous nerve is not on this map. It does not touch the bone. It runs inside the muscles in front of the humerus. That is a good answer if the professor asks why.

OBJECTIVE 3 · MEMBERS 3 AND 4

"To identify the muscles innervated by the terminal branches of the brachial plexus."

3a. Member 3 — *"Anterior compartment and upper arm flexors (e.g., musculocutaneous-innervated biceps brachii, brachialis, coracobrachialis, and primary median nerve forearm flexors)."*

ANSWER I CAN SAY ALOUD

The musculocutaneous nerve supplies all three muscles in the front of the arm: coracobrachialis, biceps brachii, and brachialis. The median nerve gives no branches in the arm. In the forearm it supplies most of the front muscles — both pronators and most wrist and finger flexors — and in the hand it supplies the thumb's thenar muscles and the two lateral lumbricals. In our patient, the three musculocutaneous muscles are the ones that lose their supply.

Muscle	Nerve	Main action
FRONT OF THE ARM — MUSCULOCUTANEOUS NERVE		

Muscle	Nerve	Main action
Coracobrachialis	Musculocutaneous	Flexes and adducts the arm at the shoulder.
Biceps brachii	Musculocutaneous	Flexes the elbow; strongest supinator; helps flex the shoulder.
Brachialis	Musculocutaneous (often a small radial twig too)	The main elbow flexor in every forearm position.
FRONT OF THE FOREARM — MEDIAN NERVE		
Pronator teres	Median	Pronation; helps flex the elbow.
Flexor carpi radialis	Median	Flexes the wrist; tilts it toward the thumb side.
Palmaris longus	Median	Weak wrist flexor (absent in some people).
Flexor digitorum superficialis	Median	Bends the fingers at the middle joints.
Flexor pollicis longus; lateral half of flexor digitorum profundus; pronator quadratus	Median (anterior interosseous branch)	Bend the thumb tip and the index and middle fingertips; pronation.
HAND — MEDIAN NERVE		
Lateral two lumbricals; opponens pollicis; abductor pollicis brevis; flexor pollicis brevis	Median	Fine finger control; opposing the thumb (touching thumb to fingertips).

3b. Member 4 — *"Posterior and lateral compartments (e.g., axillary-innervated deltoid/teres minor, radial-innervated extensors, and ulnar-innervated intrinsic hand muscles)."*

ANSWER I CAN SAY ALOUD

The axillary nerve supplies the deltoid and teres minor. The radial nerve supplies every muscle in the back of the arm and forearm: triceps, anconeus, brachioradialis, supinator, and the wrist, finger, and thumb extensors. The ulnar nerve supplies two forearm flexors — flexor carpi ulnaris and the medial half of flexor digitorum profundus — and most of the small muscles of the hand. None of these comes from the lateral cord, so in our patient they should work normally unless another nerve is also hurt.

Muscle	Nerve	Main action
Deltoid	Axillary	Abducts the arm (about 15° to 90°); front fibres flex, back fibres extend.
Teres minor	Axillary	Lateral rotation (part of the rotator cuff).
Triceps brachii; anconeus	Radial	Extend the elbow.
Brachioradialis	Radial	Flexes the elbow, best with the thumb up. It counts as a "back" muscle by nerve supply, even though it sits on the outer front of the forearm.
Extensor carpi radialis longus and brevis; extensor carpi ulnaris	Radial	Extend the wrist (hold it up).
Supinator	Radial	Supination (slow, easy turning).

Muscle	Nerve	Main action
Extensor digitorum; extensor indicis; extensor digiti minimi; thumb extensors and abductor pollicis longus	Radial	Straighten the fingers at the knuckles; move the thumb out and back.
Flexor carpi ulnaris; medial half of flexor digitorum profundus	Ulnar	Flex the wrist toward the little-finger side; bend the ring and little fingertips.
Hypothenar muscles; all interossei; medial two lumbricals; adductor pollicis	Ulnar	Little-finger movement; spreading and closing the fingers; pulling the thumb in (pinch).

SMALL WORDING POINT IN YOUR GROUP PLAN

The ulnar-supplied small hand muscles are not in the "posterior or lateral compartment". They are hand muscles. The ulnar nerve's two forearm muscles sit in the *front* compartment. The workload split is fine; only the label needs care.

OBJECTIVE 4 · MEMBERS 5 AND 6

"To discuss the functional actions affected by injuries to these muscles."

4a. Member 5 — *"Analyze shoulder abduction, external/internal rotation, and elbow flexion/extension."*

ANSWER I CAN SAY ALOUD

In our patient, elbow flexion is the main action affected. It becomes weak, not absent, because biceps and brachialis lose their nerve while brachioradialis from the radial nerve still works. Shoulder abduction and external rotation should be normal, because they depend on the axillary and suprascapular nerves, not the lateral cord — unless the axillary nerve was also stretched. Internal rotation and elbow extension should also be normal.

Action	Main muscles (nerve)	Effect in our patient
Shoulder abduction	Supraspinatus starts it, first ~15° (suprascapular n.); deltoid, ~15°–90° (axillary n.); trapezius and serratus anterior rotate the scapula above 90°.	**Normal.** Weak from about 15° upward only if the axillary nerve is also hurt.
Lateral rotation	Infraspinatus (suprascapular n.); teres minor (axillary n.).	**Normal.** Slightly weak if the axillary nerve is hurt; infraspinatus still works.
Medial rotation	Subscapularis, teres major, pectoralis major, latissimus dorsi (side branches of the plexus, not terminal branches).	**Normal.**
Shoulder flexion	Front of deltoid, pectoralis major; helped by coracobrachialis and biceps.	**Slightly weak.**
Elbow flexion	Brachialis and biceps (musculocutaneous n.); brachioradialis (radial n.).	**Weak, not lost.** Weakest with the palm up; better with the thumb up.

Action	Main muscles (nerve)	Effect in our patient
Elbow extension	Triceps, anconeus (radial n.).	**Normal.**

The suprascapular nerve and the other side branches are not terminal branches. They appear here only to explain why an action survives.

4b. Member 6 — *"Analyze forearm pronation/supination, wrist movement, and hand grip/dexterity."*

ANSWER I CAN SAY ALOUD

In our patient, supination becomes weak because the biceps, the strongest supinator, loses its nerve; the supinator muscle from the radial nerve keeps some supination. Pronation, wrist movement, and grip should be largely normal. If the lateral root of the median nerve is also hurt, pronation and wrist flexion become weaker. Fine finger control should be preserved, because the small hand muscles get their fibres through the medial cord.

Action	Main muscles (nerve)	Effect in our patient
Supination	Biceps — strongest, especially with the elbow bent (musculocutaneous n.); supinator (radial n.).	**Weak**, most of all with the elbow bent.
Pronation	Pronator teres, pronator quadratus (median n.).	**Normal**; weaker if the lateral root of the median is hurt.
Wrist flexion	Flexor carpi radialis, palmaris longus (median n.); flexor carpi ulnaris (ulnar n.).	**Normal**; if the lateral root is hurt, weaker and pulled toward the little-finger side.
Wrist extension	Radial and ulnar wrist extensors (radial n.).	**Normal.**
Power grip	Long finger flexors (median and ulnar n.), with wrist extensors holding the wrist up (radial n.).	**Normal.** Note: a *radial* injury weakens grip, because a dropped wrist makes the finger flexors work poorly.
Fine dexterity	Thenar muscles (median n.); interossei and adductor pollicis (ulnar n.); lumbricals (both).	**Normal.**

OBJECTIVE 5 · MEMBERS 7 AND 8

"To describe the specific deficits associated with each nerve injury."

Two terms: *anaesthesia* means complete loss of feeling in a skin area. *Paraesthesia* means tingling or "pins and needles".

5a. Member 7 — *"Detail the clinical presentation of Musculocutaneous, Axillary, and Median nerve lesions (e.g., loss of elbow flexion, 'badge' area anesthesia over the deltoid, hand posture changes)."*

ANSWER I CAN SAY ALOUD

A musculocutaneous lesion — our patient's most likely injury — gives weak elbow flexion and supination, a weak biceps reflex, and numbness on the thumb side of the forearm. An axillary lesion gives weak abduction and lateral rotation, a flattened shoulder once the deltoid wastes, and numbness over the "regimental badge" patch on the outer shoulder. A high median lesion gives weak pronation and wrist flexion, inability to bend the index and middle fingers when making a fist, weak thumb opposition with thenar wasting, and numbness of the thumb-side palm and 3½ fingers.

Nerve	Motor loss	Sign or posture	Sensory loss
Musculocutaneous our patient	Weak elbow flexion and supination; slightly weak shoulder flexion; weak biceps reflex.	No classic deformity. Later, the front of the arm wastes.	Thumb side of the forearm.
Axillary	Weak abduction above ~15° (deltoid); weak lateral rotation (teres minor).	Flattened, "squared" shoulder once the deltoid wastes.	Skin over the lower deltoid — the "regimental badge" area.
Median (high lesion, at or above the elbow)	Weak pronation and wrist flexion; cannot bend the thumb tip or the index and middle fingers; weak opposition.	**Hand of benediction** when trying to make a fist (index and middle fingers stay straight); thenar wasting with the thumb lying flat ("ape hand").	Thumb-side palm; palm side of the thumb, index, middle, and half the ring finger.

NOTE ON YOUR GROUP PLAN

The breakdown says "loss of elbow flexion" for a musculocutaneous lesion. In practice it is **weakness**, not loss. Brachioradialis (radial nerve) and, in many people, a radial twig to brachialis still flex the elbow. Say "weak elbow flexion" in class.

CLASSIC SIGN VERSUS REAL PATIENTS

The "hand of benediction" is the classic textbook sign of a high median lesion, and exams ask it in that form. A 2022 case series of high median nerve lesions found that none of its patients showed the sign on examination. Learn the classic sign for the exam. Remember that a real patient may not show it.

5b. Member 8 — *"Detail Radial and Ulnar nerve lesions (e.g., wrist drop, claw hand deformity) along with general sensory mapping patterns."*

ANSWER I CAN SAY ALOUD

A radial lesion at the radial groove gives wrist drop: the patient cannot lift the wrist or straighten the fingers at the knuckles, grip is weak because the wrist hangs down, and the back of the hand near the thumb is numb; the triceps is usually spared at this level. An ulnar lesion gives a claw hand of the ring and little fingers, weak finger spreading, hypothenar wasting, and numbness of the little finger and half the ring finger. Neither nerve comes from the lateral cord, so our patient should not show these signs.

Nerve	Motor loss	Sign or posture	Sensory loss
Radial (at the radial groove)	Weak wrist and finger extension; weak brachioradialis; weak supination with the elbow straight. Triceps usually spared (its branches leave higher up).	**Wrist drop.** Weak grip, because the wrist cannot be held up.	Back of the hand near the thumb (first web space).
Ulnar (at the elbow)	Weak finger spreading and closing (interossei); weak pinch (adductor pollicis); weak little-finger movement.	**Claw hand** of the ring and little fingers: knuckles over-straightened, finger joints bent. Wasted hypothenar eminence.	Little finger and half the ring finger, front and back.

Why the claw forms. The lumbricals and interossei normally bend the knuckles and straighten the finger joints. When they are weak, the long extensors over-straighten the knuckles and the long flexors bend the finger joints. A lesion high at the elbow also weakens the ulnar half of flexor digitorum profundus, so the clawing is often *milder* than in a lesion at the wrist. This is called the "ulnar paradox".

The sensory map — one test spot per nerve

Nerve	Skin area	Quick test spot
Musculocutaneous	Thumb side of the forearm	Outer forearm, just below the elbow crease
Axillary	Lower outer shoulder (badge area)	Skin over the lower deltoid
Radial	Back of the arm, forearm, and thumb-side hand	Back of the first web space (between thumb and index)
Median	Thumb-side palm and 3½ fingers	Pad of the index finger
Ulnar	Little finger and half the ring finger	Pad of the little finger

6. Likely professor follow-up questions

F1. Why did the shoulder dislocate forward and not backward?

The arm was out to the side and turned outward. In that position the head already presses on the front of the socket, and the front-lower capsule is the tightest and weakest support. A side blow levers the head forward over the front-lower rim. A backward (posterior) dislocation needs the opposite position: arm in and turned inward.

F2. The lateral cord has two terminal branches. Why do you pick the musculocutaneous nerve?

Three reasons. It is the lateral cord's only complete, named terminal nerve; the other branch is just one root of the median. The case says "its terminal branch" in the singular. And the musculocutaneous nerve is held fairly fixed where it passes through coracobrachialis, so a head pushing forward stretches it. I would still check pronation and wrist flexion, because the lateral root of the median can be hurt too.

F3. Which nerve is most often injured in anterior dislocation? How would you test it, and when?

The axillary nerve. It wraps around the surgical neck, so a head that drops forward and down pulls it tight. Test deltoid strength (resisted abduction) and light touch over the lower deltoid. Do this before reduction and again after it, so that you know whether a deficit was caused by the injury or by the reduction.

F4. The musculocutaneous nerve is damaged. Why can he still bend his elbow?

Brachioradialis, supplied by the radial nerve, still flexes the elbow — best with the thumb pointing up. Many people also have a small radial branch to the outer brachialis. So flexion is weak, not lost. This is why a good examination tests flexion with the palm up, which brings out the biceps weakness.

F5. How do you know the weakness comes from a nerve, and not from pain or a torn biceps?

Look at the skin. A nerve injury causes numbness in that nerve's own skin patch — here, the thumb side of the forearm. A torn biceps tendon or pain causes weakness with normal skin. A weak or absent biceps reflex also points to the nerve. Finally, the pattern matters: the weak muscles all belong to one nerve.

F6. In the ER the shoulder looks "squared off". Is that axillary nerve damage?

Not by itself. Right after an anterior dislocation the head has left its normal place under the deltoid, so the bony tip of the shoulder (the acromion) stands out and the shoulder looks square. That shape appears immediately. A flat shoulder from axillary nerve damage comes from deltoid *wasting*, which takes weeks. The shapes look alike, but the cause and the timing differ.

F7. What if his arm had been pulled in and turned inward at the moment of injury?

Then a posterior dislocation would be more likely. The head would move backward, away from the cords in front of the joint. So this case's mechanism — the head pushing the cords forward — would not apply in the same way.

F8. Common mistake: "The median nerve is a lateral cord nerve, so the whole median nerve is lost here." Is that right?

No. The median nerve has two roots: one from the lateral cord and one from the medial cord. A lateral cord injury can affect only the lateral-root fibres. These go mainly to the forearm flexors and pronator teres. The thumb (thenar) muscles get their fibres from the medial root, so they are usually spared.

7. Quick recap and self-check

The case in my own words

A young rugby player was tackled while his arm was out to the side and turned outward. That position levered the head of his humerus forward out of the shallow socket, and it came to rest under the coracoid. That space is where the cords of the brachial plexus run, so the head pushed the lateral cord forward and stretched it. The lateral cord's own terminal nerve, the musculocutaneous nerve, was damaged. Its three muscles — coracobrachialis, biceps, and brachialis — lost their signal, so elbow flexion and supination became weak, and the thumb side of his forearm went numb. He can still bend his elbow weakly, because brachioradialis from the radial nerve still works. The axillary nerve should also be checked, because it is the nerve most often hurt in this injury.

Essential takeaways

1. **Objective 1.** Five terminal branches: musculocutaneous, axillary, radial, median, ulnar. The lateral cord gives the musculocutaneous nerve and the lateral root of the median nerve.
2. **Objective 2.** The humeral head moves forward and down under the coracoid and presses on the cords. The surgical neck carries the axillary nerve. The radial groove, the supracondylar region, and the medial epicondyle carry the radial, median, and ulnar nerves.
3. **Objective 3.** The musculocutaneous nerve supplies coracobrachialis, biceps brachii, and brachialis, plus the skin of the thumb side of the forearm.
4. **Objective 4.** Elbow flexion and supination become weak, not lost, because radial-nerve muscles (brachioradialis, supinator) still help.
5. **Objective 5.** Musculocutaneous lesion: weak elbow flexion and supination, a weak biceps reflex, and a numb lateral forearm. Each of the other four nerves has its own signature: badge-area numbness, hand of benediction, wrist drop, and claw hand.
6. **Real-world habit.** In any anterior dislocation, test the axillary nerve (deltoid strength and badge-area skin) before and after reduction.

COMMON MEMORY AIDS — NOT FROM YOUR LECTURE

- **"Randy Travis Drinks Cold Beer"** — the order of the plexus: Roots, Trunks, Divisions, Cords, Branches. Use it when you trace the lateral cord back to C5–C7.
- **"BBC"** — the musculocutaneous nerve supplies Biceps, Brachialis, and Coracobrachialis. These are exactly the muscles our patient loses.
- **"LOAF"** — the median nerve's hand muscles: Lateral two lumbricals, Opponens pollicis, Abductor pollicis brevis, Flexor pollicis brevis. These are the muscles that stay *normal* in a lateral cord injury, because their fibres use the medial root.

Self-check — answer first, then turn the page

1. Name the two terminal branches of the lateral cord.
2. Where does the head usually come to rest in an anterior dislocation, and why does that matter for the cords?
3. After a musculocutaneous injury, a patient can still bend the elbow weakly with the thumb pointing up. Which muscle allows this, and which nerve supplies it?
4. Which skin spot would you test for the musculocutaneous nerve? Which for the axillary nerve?
5. Surgical neck or anatomical neck — which one has the axillary nerve wrapped around it?
6. After a dislocation, a patient has weak elbow flexion, but forearm sensation and the biceps reflex are normal. What should you consider, and why?

Self-check answers

1. The **musculocutaneous nerve** and the **lateral root of the median nerve**.
2. **In front of and below the socket, under the coracoid process** (subcoracoid). The cords pass through that same space behind pectoralis minor, so the head pushes them forward and stretches them.
3. **Brachioradialis**, supplied by the **radial nerve**. It flexes the elbow best in the thumb-up position.
4. Musculocutaneous: the **thumb side of the forearm**, just below the elbow. Axillary: the **skin over the lower deltoid** (badge area).
5. The **surgical neck**. The anatomical neck is the groove at the edge of the head.
6. A nerve injury would usually make the skin numb and weaken the reflex, so a pure nerve cause is less likely. Consider a **biceps muscle or tendon injury**, or **pain limiting effort**. Re-examine after reduction and pain control.

Appendix A. Source check

HOW THESE SOURCES WERE CHECKED — PLEASE READ

The network policy of my work environment blocked direct access to NCBI Bookshelf, PubMed Central, StatPearls, and most anatomy sites. I verified each claim through web-search results that display text from the source. The excerpts below are copied exactly as those search results showed them. I could **not** open the full pages. So I could not confirm page numbers or section headings, and I could not check each excerpt against the live page. Open each link once before you quote it in class.

#	Claim checked	Excerpt (as shown in the search result)	Source and locator
1	Anterior dislocation is the most common type; the usual mechanism is a blow to an abducted, externally rotated, extended arm.	*"Anterior dislocation is the most common, accounting for up to 97% of all shoulder dislocations. Mechanism of injury is usually a blow to an abducted, externally rotated and extended extremity."*	Abrams & Akbarnia, *Shoulder Dislocations Overview*, StatPearls. ncbi.nlm.nih.gov/books/NBK459125. Section: not confirmed.
2	Peak age 15–30; more common in males.	*"the highest incidence of anterior shoulder dislocation is seen between the ages of 15 to 30, with a male to female ratio of 2.64."*	StatPearls continuing-education page on anterior shoulder dislocation. statpearls.com/sp/al/48/86948. Exact article title not shown in the result.
3	Hill–Sachs lesion: a dent on the back-outer humeral head from impaction against the glenoid.	*"Cortical depression of posterolateral head of the humerus related to impaction of the humeral head with inferior glenoid in anterior shoulder dislocation."*	LITFL, *Hill-Sachs defect*. litfl.com/hill-sachs-lesion. Definition line.
4	The cords are named by their position around the second part of the axillary artery.	*"named as per their relationship with the second part as lateral, medial, and posterior cords."*	Sharma et al. (2024), *Cureus*. PMC11255909. Section: not confirmed.
5	The musculocutaneous nerve supplies coracobrachialis, biceps brachii, and brachialis.	*"It innervates the three muscles of the anterior compartment of the arm"*	*Anatomy, Shoulder and Upper Limb, Musculocutaneous Nerve*, StatPearls. NBK534199; PubMed 30480938. The search result named the three muscles and stated the nerve is "a terminal branch of the lateral cord".
6	A lateral cord injury weakens elbow flexion (musculocutaneous) and wrist flexion (lateral root of median).	*"injury to lateral cord impairs elbow flexion (musculocutaneous) and wrist flexion (lateral root of median nerve)"*	Wheeless' Textbook of Orthopaedics, *Lateral cord*. wheelessonline.com/nerves/lateral-cord.

#	Claim checked	Excerpt (as shown in the search result)	Source and locator
7	The musculocutaneous nerve can be stretched where it is fixed in coracobrachialis (a proposed mechanism).	*"The mechanism of injury is speculated to be a sudden overloading of the anterior shoulder with extension and external rotation, which overstretched and compressed the MCN within the coracobrachialis muscle where the nerve is relatively fixed."*	Kwon et al. (2019), *Neurology Asia*, 24(3), 267–270. Yonsei repository record.
8	Nerve injury is common after shoulder dislocation and humeral neck fracture; the musculocutaneous nerve is among those involved.	*"45%, most involving the axillary, suprascapular, radial and musculocutaneous nerves"*	de Laat et al. (1994), *JBJS Br*, 76(3), 381–383. Abstract. orthobullets.com/evidence/8175837 (abstract copy); PubMed 8175837.
9	The axillary nerve is the nerve most often involved in anterior dislocation.	*"the axillary nerve was most frequently involved (42%)"*	Visser et al. (1999), *JBJS Br*, 81(4), 679–685. Abstract finding as shown in the search result. boneandjoint.org.uk; PubMed 10463745.
10	The axillary nerve is injured by surgical-neck trauma or anterior dislocation.	*"most commonly injured by trauma to the surgical neck of the humerus or anterior shoulder dislocations"*	LITFL, *Axillary nerve lesions*. litfl.com/axillary-nerve-lesions.
11	The axillary nerve's skin branch supplies the lower outer deltoid (badge area).	*"the upper lateral cutaneous branch of the posterior axillary division, which follows teres minor innervation, supplies the lateral inferior deltoid skin."*	StatPearls entry on the axillary nerve. statpearls.com/point-of-care/18058. Title and section not confirmed.
12	Supraspinatus starts abduction (first 15°); trapezius and serratus anterior act above 90°.	*"is responsible for the initiation of arm abduction by stabilizing the humeral head in the glenoid fossa and is in control of the motion up to the first 15 degrees of abduction." / "The trapezius and serratus anterior coordinate with each other and the scapula to facilitate abduction of the arm upwards of 90 degrees."*	*Anatomy, Shoulder and Upper Limb, Arm Abductor Muscles*, StatPearls. NBK537148.
13	Median nerve territory; hand of benediction occurs only with proximal lesions.	*"motor innervation to the anterior forearm flexors, the thenar muscles, and the two lateral lumbricals as well as sensory innervation to the lateral palm and anterior, lateral three and a half fingers" / "the hand of benediction only occurs in proximal median nerve injuries"*	AMBOSS, *Median nerve neuropathy*. amboss.com.
14	Real patients with high median lesions may not show the hand of benediction.	*"none of the patients presented in this study had a hand of benediction on physical examination despite the presence of a high median nerve lesion."*	*Plastic and Reconstructive Surgery – Global Open* (Oct 2022), case series. DOAJ record. Authors not shown in the result.
15	Humeral shaft fractures in the middle–distal region are linked to radial nerve injury.	*"Transverse spiral fractures between the middle and distal parts of the humerus are more likely to be associated with radial nerve injury."*	Medscape, *Radial Mononeuropathy*. Medscape app page.
16	The ulnar nerve comes from the medial cord; claw hand posture.	*"the ulnar nerve is the terminal branch of the medial cord (C8, T1)" / "metacarpophalangeal (MCP) hyperextension and proximal interphalangeal (PIP) and distal interphalangeal (DIP) flexion"*	*Claw Hand*, StatPearls (text shown on a CE course listing). NBK536922; cebroker.com listing.

Claims supported only by summaries, or not checked against an excerpt

These are standard textbook teaching. I could not inspect an exact excerpt for them. Check them against your lecture or Moore's *Clinically Oriented Anatomy*.

Claim	Status
Brachialis often gets a small radial nerve branch.	Supported by study summaries (Ilayperuma et al., 2019, reported a radial contribution in most of 240 limbs; other studies 67–90%). Exact abstract wording not inspected.
Brachioradialis (radial nerve) flexes the elbow, best with the thumb up.	Supported by anatomy-site summaries. No authoritative excerpt inspected.
Biceps is the strongest supinator, especially with the elbow bent.	Standard textbook teaching. Not checked against an excerpt.
Triceps is usually spared in a mid-shaft radial nerve injury.	Supported by summaries of StatPearls *Wrist Drop* and Medscape. One search did not find triceps wording in StatPearls. Partly verified.
The lateral root of the median nerve carries sensory fibres for the thumb, index, and middle fingers.	Supported only by a study-aid source and an electrodiagnosis reference summary. Marked "probably" in the text.
Supracondylar fractures can injure the median nerve.	Standard textbook teaching. Not checked against an excerpt.
Ulnar paradox (milder clawing with a high lesion).	Supported by a lecture-site summary (Lecturio). No authoritative excerpt inspected.
The quick sensory test spots for each nerve.	Standard clinical teaching. Not checked against an excerpt.
Author initials and update dates of StatPearls entries.	Not confirmed. Where I could not confirm authors, the reference is listed by title.

Appendix B. References (APA 7)

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Recommended textbook for checking the unverified items: Moore, K. L., Dalley, A. F., & Agur, A. M. R. *Clinically oriented anatomy* (Wolters Kluwer). Edition and page numbers not given, because I could not inspect a copy.